Supplement to "Silent Aliasing in Fixed Fourier-Feature Coordinate Models": Full Proofs

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Numbering follows the main paper (T1–T4 are Theorems 1–4 there). Setting and notation as in the main paper: $e_\omega(x) = e^{i2\pi\omega x}$ on $[0, 1)$, model set $\Lambda = \{\omega_1, \ldots, \omega_m\}$, samples $T = \{t_j\}_{j=1}^N$, synthesis matrix $\boldsymbol{\Phi}_{jk} = e_{\omega_k}(t_j)$, sample atom $\boldsymbol{\phi}_\nu = (e^{i2\pi\nu t_j})_j$, visibility $v_T(\nu) = \|(\boldsymbol{I} - \boldsymbol{P}_\Lambda)\boldsymbol{\phi}_\nu\|/\sqrt{N}$, aliasability $a_T(\nu) = \|\boldsymbol{\Phi}^\dagger \boldsymbol{\phi}_\nu\|$. Every constant below is also asserted numerically in `tests/test_identifiability.py`.

1 Proof of Theorem 1 (exact equivalence on grid subsets)

(i) $v_T(\nu) = 0$ iff $(\boldsymbol{I} - \boldsymbol{P}_\Lambda)\boldsymbol{\phi}_\nu = 0$ iff $\boldsymbol{\phi}_\nu \in \operatorname{range} \boldsymbol{\Phi}$, which is precisely the statement that some coefficient vector $\boldsymbol{c}$ has $\boldsymbol{\Phi c} = \boldsymbol{\phi}_\nu$, i.e. the unit tone at ν agrees with a model element at every sample. For the grid statement, let $t_j = g_j/Q$ with $g_j \in \{0, \ldots, Q-1\}$ and $\nu = \omega_k + rQ$, $r \in \mathbb{Z}$. Then

$$e^{i2\pi\nu g_j/Q} = e^{i2\pi\omega_k g_j/Q}\, e^{i2\pi r g_j} = e^{i2\pi\omega_k g_j/Q},$$

since $rg_j \in \mathbb{Z}$; hence $\boldsymbol{\phi}_\nu = \boldsymbol{\phi}_{\omega_k}$ entrywise, for *every* subset of the grid. Consequently $v_T(\nu) = 0$; moreover, because $\sigma_{\min}(\boldsymbol{\Phi}) > 0$ makes the LS solution unique, the fit of $y = \boldsymbol{\phi}_\nu$ is the unique $\boldsymbol{c}$ with $\boldsymbol{\Phi c} = \boldsymbol{\phi}_{\omega_k}$, namely the k-th unit vector, and the residual is exactly zero.

(ii) By (i), $S_T f_1 = S_T f_2$ pointwise. For any noise mechanism that acts on $S_T f$ (additive or otherwise, as long as its law depends on the signal only through $S_T f$), the observation distributions under f_1 and f_2 are therefore identical. Let $\hat{f} = \delta(\boldsymbol{y})$ be any estimator. For every realization $\boldsymbol{y}$,

$$\|\delta(\boldsymbol{y}) - f_1\|_{L^2} + \|\delta(\boldsymbol{y}) - f_2\|_{L^2} \;\geq\; \|f_1 - f_2\|_{L^2} = |a|\,\|e_\nu - e_{\omega_k}\|_{L^2} = |a|\sqrt{2},$$

using the triangle inequality and, for the last equality, that e_ν and e_{ω_k} are distinct integer frequencies, hence orthonormal in $L^2[0, 1)$. Thus $\max_i \|\delta(\boldsymbol{y}) - f_i\| \geq |a|/\sqrt{2}$ pointwise (a two-point bound in the sense of Le Cam, here even realization-wise).

(iii) If $[\boldsymbol{\Phi}\ \boldsymbol{\phi}_\nu]$ has full column rank then $\boldsymbol{\phi}_\nu \notin \operatorname{range} \boldsymbol{\Phi}$, so $v_T(\nu) > 0$ by (i). □

Remark 1. *The full-column-rank hypothesis on $\boldsymbol{\Phi}$ requires $N \geq m$ and the elements of Λ to be distinct modulo Q (two atoms congruent mod Q have identical columns on any grid subset). "Any N" claims are always to be read as "any grid-subset size N for which this rank condition holds".*

2 Proof of Theorem 2 (Riesz conversion and decomposition)

(i) G_{L^2} is the Gram matrix of $\{e_{\omega_k}\}$ in $L^2[0, 1)$: $\|\sum_k c_k e_{\omega_k}\|_{L^2}^2 = \boldsymbol{c}^* G_{L^2} \boldsymbol{c}$, and the Rayleigh quotient of the Hermitian PSD matrix G_{L^2} lies in $[\lambda_{\min}, \lambda_{\max}]$. The entries are $\int_0^1 e^{i2\pi(\omega_\ell - \omega_k)t}\, dt = \frac{e^{i2\pi\delta} - 1}{i2\pi\delta} = e^{i\pi\delta}\, \frac{\sin \pi\delta}{\pi\delta}$ for $\delta = \omega_\ell - \omega_k \neq 0$, and 1 on the diagonal. G_{L^2} is singular iff the atoms are linearly dependent in L^2, which for finitely many distinct frequencies never happens (distinct exponentials are linearly independent); hence $\lambda_{\min} > 0$ for distinct frequencies.

(ii) Expand $\hat{f} - f = \sum_k \Delta c_k e_{\omega_k} - \sum_l a_l e_{\nu_l}$ and evaluate $\|\hat{f} - f\|_{L^2}^2$ as the quadratic form of the joint Gram matrix of $(\{\omega_k\}, \{\nu_l\})$ applied to $(\Delta \boldsymbol{c}, -\boldsymbol{a})$; grouping blocks gives Eq. (3) of the main paper. For all-integer frequencies the off-diagonal blocks vanish (orthonormality), killing the cross term. □

3 Proof of Theorem 3 (random sampling breaks exact aliasing)

We use the following complex Hoeffding bound throughout: if $Z_1, \ldots, Z_N$ are independent with $|Z_j| \le 1$, then for $\bar{Z} = \frac{1}{N}\sum_j Z_j$,

$$\Pr\left(|\bar{Z} - \mathbb{E}\bar{Z}| \ge \varepsilon\right) \;\le\; 4\exp(-N\varepsilon^2/4), \tag{1}$$

obtained by applying the real Hoeffding inequality to the real and imaginary parts (each in $[-1, 1]$) with radius $\varepsilon/\sqrt{2}$ and a union bound.

3.1 Part (a): jitter

Let $t_j = g_j/Q + \eta_j$ with g_j grid points and η_j i.i.d. mean-zero with characteristic function χ_η, and $\nu - \omega_k = rQ$, $r \in \mathbb{Z} \setminus \{0\}$. The fold coherence is

$$C = \frac{1}{N}\sum_j e^{i2\pi(\nu-\omega_k)t_j} = \frac{1}{N}\sum_j \underbrace{e^{i2\pi r g_j}}_{=1} e^{i2\pi rQ\eta_j} = \frac{1}{N}\sum_j e^{i2\pi rQ\eta_j},$$

an empirical characteristic function with $\mathbb{E}C = \chi_\eta(2\pi rQ)$ and concentration (1): $|C - \chi_\eta(2\pi rQ)| \le 2\sqrt{\log(4/\delta)/N}$ with probability $\ge 1-\delta$. For Gaussian jitter, $\chi_\eta(u) = e^{-u^2\sigma_t^2/2}$, so $\chi_\eta(2\pi rQ) = e^{-2(\pi rQ\sigma_t)^2}$; for uniform jitter of width w, $\chi_\eta(2\pi rQ) = \operatorname{sinc}(rQw)$.

From coherence to visibility. Let $\boldsymbol{b} = \boldsymbol{\Phi}^*\boldsymbol{\phi}_\nu/N \in \mathbb{C}^m$ and $\boldsymbol{G}_N = \boldsymbol{\Phi}^*\boldsymbol{\Phi}/N$. Then

$$v_T(\nu)^2 = \frac{1}{N}\|\boldsymbol{\phi}_\nu\|^2 - \frac{1}{N}\boldsymbol{\phi}_\nu^*\boldsymbol{P}_\Lambda\boldsymbol{\phi}_\nu = 1 - \boldsymbol{b}^*\boldsymbol{G}_N^{-1}\boldsymbol{b}.$$

Assume additionally—as in the theorem statement—that the grid points g_j are drawn i.i.d. uniformly from the grid (a uniformly random subset drawn without replacement obeys the same bounds with the same constants by Serfling's inequality for sampling without replacement) and that the elements of Λ are distinct modulo Q (otherwise two Gram columns coincide on every grid subset and $\boldsymbol{G}_\infty \ne \boldsymbol{I}$, breaking the perturbation step below). Each entry of $\boldsymbol{b}$ other than the fold entry, and each off-diagonal entry of $\boldsymbol{G}_N$, is an empirical mean of independent unit-modulus terms whose expectation vanishes (distinct residues modulo Q average to zero over the uniform grid, and the jitter factor only multiplies by a number of modulus ≤ 1); the fold entry of $\boldsymbol{b}$ is C above. By (1) and a union bound over these $m + m^2$ quantities, with probability $\ge 1-\delta$ all of them are within $\varepsilon = 2\sqrt{\log(4(m+m^2)/\delta)/N}$ of their means. Writing $\boldsymbol{G}_N = \boldsymbol{I} + \boldsymbol{E}$ with $\|\boldsymbol{E}\| \le m\varepsilon$ and $\boldsymbol{b} = C\boldsymbol{e}_k + \boldsymbol{u}$ with $\|\boldsymbol{u}\| \le \sqrt{m}\,\varepsilon$, a first-order expansion of $\boldsymbol{b}^*\boldsymbol{G}_N^{-1}\boldsymbol{b}$ gives

$$\left|v_T(\nu)^2 - \left(1 - |\chi_\eta(2\pi rQ)|^2\right)\right| \;\le\; c\,m\,\varepsilon$$

for an absolute constant c (valid while $m\varepsilon < 1/2$), which is the $O_P\big(m\sqrt{\log(m/\delta)/N}\big)$ term of the theorem. The small-jitter law follows from $1 - e^{-2u} \le 2u$ and $1 - e^{-2u} \to 2u$ as $u \to 0$ with $u = (2\pi rQ\sigma_t)^2/2$: $v_T(\nu) = \sqrt{1-\chi^2} + O(m\varepsilon) = 2\pi|r|Q\sigma_t\,(1 + o(1))$.

3.2 Part (b): i.i.d. samples

Entries of $\boldsymbol{G}_N - \boldsymbol{G}_\infty$ and of $\boldsymbol{b}_\nu = \boldsymbol{\Phi}^*\boldsymbol{\phi}_\nu/N$ (for every $\nu \in \Omega$) are empirical means of independent unit-modulus variables; by (1) and a union bound over the $m^2 + mK$ of them, with probability $\ge 1-\delta$ every such quantity is within $\varepsilon = 2\sqrt{\log(4(mK + m^2)/\delta)/N}$ of its limit. Under the hypotheses, the limiting coherences vanish, so $\|\boldsymbol{b}_\nu\| \le \sqrt{m}\,\varepsilon$ for all $\nu \in \Omega$, and $\|\boldsymbol{G}_N - \boldsymbol{G}_\infty\| \le \|\cdot\|_F \le m\varepsilon$, hence $\lambda_{\min}(\boldsymbol{G}_N) \ge \lambda_{\min} - m\varepsilon > 0$. Therefore, uniformly over Ω,

$$a_T(\nu) = \|\boldsymbol{G}_N^{-1}\boldsymbol{b}_\nu\| \le \frac{\|\boldsymbol{b}_\nu\|}{\lambda_{\min}(\boldsymbol{G}_N)} \le \frac{\sqrt{m}\,\varepsilon}{\lambda_{\min} - m\varepsilon}. \qquad \square$$

Remark 2. *The bound is a union bound and is loose by design (Fig. 1b of the main paper); its role is the rate $N^{-1/2}$ and the uniformity over the candidate set, both matched empirically. It is finite exactly when $m\varepsilon < \lambda_{\min}$ and is reported as vacuous otherwise, never silently extrapolated.*

4 Proof of Theorem 4 (detection dichotomy)

(a) If $S_T f_1 = S_T f_2$ and the observation law depends on the signal only through $S_T f$ (as for additive noise $\boldsymbol{y} = S_T f + \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon}$ independent of f), the distributions of $\boldsymbol{y}$ under the two hypotheses are equal. For any (possibly randomized) test $\varphi(\boldsymbol{y}) \in [0,1]$, $\mathbb{E}_{f_1}\varphi = \mathbb{E}_{f_2}\varphi$: power equals size.

(b) Under $\boldsymbol{y} = \boldsymbol{D}\boldsymbol{\beta} + \boldsymbol{s} + \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim \mathcal{N}(0, \sigma^2 \boldsymbol{I})$ and $\boldsymbol{P}_D$ the projector onto the rank-p column space of $\boldsymbol{D}$, $(\boldsymbol{I} - \boldsymbol{P}_D)\boldsymbol{y} = (\boldsymbol{I} - \boldsymbol{P}_D)(\boldsymbol{s} + \boldsymbol{\varepsilon})$ is Gaussian with mean $(\boldsymbol{I} - \boldsymbol{P}_D)\boldsymbol{s}$ and covariance $\sigma^2(\boldsymbol{I} - \boldsymbol{P}_D)$; hence $\|(\boldsymbol{I} - \boldsymbol{P}_D)\boldsymbol{y}\|^2/\sigma^2$ is noncentral χ^2 with $N - p$ degrees of freedom and noncentrality $\lambda = \|(\boldsymbol{I} - \boldsymbol{P}_D)\boldsymbol{s}\|^2/\sigma^2$ (central when $\boldsymbol{s} = 0$ or when the component is coherent, $(\boldsymbol{I} - \boldsymbol{P}_D)\boldsymbol{s} = 0$). The stated power expression is the definition of the noncentral-χ^2 survival function at the central $(1-\alpha)$-quantile. □

5 Background: average-case decomposition

Proposition S 1 (average-case LS error; standard bookkeeping)**.** *Let $\boldsymbol{y} = \boldsymbol{\Phi}\boldsymbol{c}^\star + \boldsymbol{\Phi}_{\text{out}}\boldsymbol{a} + \boldsymbol{\varepsilon}$ with $\sigma_{\min}(\boldsymbol{\Phi}) > 0$, the a_l zero-mean, uncorrelated, $\mathbb{E}|a_l|^2 = p_l$, independent of $\boldsymbol{\varepsilon}$, and $\mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^*] = \sigma^2\boldsymbol{I}$. Then, with $\boldsymbol{M} = (\boldsymbol{\Phi}^*\boldsymbol{\Phi})^{-1}\boldsymbol{\Phi}^*$,*

$$\mathbb{E}\|\hat{\boldsymbol{c}} - \boldsymbol{c}^\star\|^2 = \sigma^2 \operatorname{tr}\big((\boldsymbol{\Phi}^*\boldsymbol{\Phi})^{-1}\big) + \operatorname{tr}\big(\boldsymbol{M}\,\boldsymbol{\Phi}_{\text{out}}\operatorname{diag}(\boldsymbol{p})\boldsymbol{\Phi}_{\text{out}}^*\,\boldsymbol{M}^*\big).$$

Proof. $\hat{\boldsymbol{c}} - \boldsymbol{c}^\star = \boldsymbol{M}(\boldsymbol{\Phi}_{\text{out}}\boldsymbol{a} + \boldsymbol{\varepsilon})$; expand the squared norm; the cross term vanishes by independence and zero means; the two quadratic terms are the stated traces. □

This proposition is retained for completeness only; it is standard omitted-variable bias/variance bookkeeping and is not claimed as a contribution.

6 Sampling design: surrogate and the continuum certificate

Relation to classical design. Write $\boldsymbol{\Psi} = \boldsymbol{\Phi}_\Omega$ for the atoms of the candidate out-of-band set Ω. The design goal—keep $\frac{1}{N}\boldsymbol{\Phi}^*\boldsymbol{\Phi}$ near $\boldsymbol{I}$ while driving $\frac{1}{N}\boldsymbol{\Phi}^*\boldsymbol{\Psi}$ toward 0, indifferent to $\frac{1}{N}\boldsymbol{\Psi}^*\boldsymbol{\Psi}$—is the D_s-optimality (interest-vs-nuisance) criterion of experimental design and LCMV/MVDR null-steering: the information about the interest block is the Schur complement $J_{\Lambda\Lambda} - J_{\Lambda\Omega}J_{\Omega\Omega}^{-1}J_{\Omega\Lambda}$, which equals the isolated interest block iff the cross-block $J_{\Lambda\Omega} = \boldsymbol{\Phi}^*\boldsymbol{\Psi}$ vanishes. We do not claim this principle; our additions are the identifiability reading (Thm. 1 says grid designs *cannot* realize it for coherent folds) and the certificate below.

Proposition S 2 (surrogate controls aliasability)**.** *Let $\mu(T) = \max_{k,\nu} |(\boldsymbol{\Phi}^*\boldsymbol{\Psi})_{k\nu}|/N$ and let $\lambda_0 = \lambda_{\min}(\boldsymbol{\Phi}^*\boldsymbol{\Phi}/N) > 0$. Then for every $\nu \in \Omega$, $a_T(\nu) = \|(\boldsymbol{\Phi}^*\boldsymbol{\Phi}/N)^{-1}(\boldsymbol{\Phi}^*\boldsymbol{\phi}_\nu/N)\| \le \sqrt{m}\,\mu(T)/\lambda_0$, and $\mu(T)^2 \le N^2 E_{\Lambda\Omega}(T)$ where $E_{\Lambda\Omega}$ is the cross-energy term of the surrogate J. Hence minimizing J (which also drives $\lambda_0 \to 1$ through $E_{\Lambda\Lambda}$) upper-controls the worst-case aliasability over Ω.*

Proof. $\boldsymbol{\Phi}^*\boldsymbol{\phi}_\nu/N$ is the ν-column of $\boldsymbol{\Phi}^*\boldsymbol{\Psi}/N$, of norm $\le \sqrt{m}\,\mu(T)$; multiply by $\|(\boldsymbol{\Phi}^*\boldsymbol{\Phi}/N)^{-1}\| = 1/\lambda_0$. The μ–E inequality is max $\le \ell_2$. No submodularity is claimed: a_T depends on $\boldsymbol{\Phi}^\dagger$ nonlinearly, so the greedy/coordinate procedures are heuristics with deterministic outputs, not approximation-ratio algorithms. □

Proposition S 3 (continuum certificate; Prop. 3 in the paper)**.** *Fix T with $\sigma_{\min}(\boldsymbol{\Phi}) > 0$ and let $\boldsymbol{M} = \boldsymbol{\Phi}(\boldsymbol{\Phi}^*\boldsymbol{\Phi})^{-2}\boldsymbol{\Phi}^*$ (Hermitian PSD). Then $g(\nu) := a_T(\nu)^2 = \boldsymbol{\phi}_\nu^*\boldsymbol{M}\boldsymbol{\phi}_\nu = \sum_{j,l}\boldsymbol{M}_{jl}\,e^{i2\pi\nu(t_l - t_j)}$ is a real trigonometric polynomial in ν, L-Lipschitz with $L = 2\pi\sum_{j,l}|\boldsymbol{M}_{jl}|\,|t_l - t_j|$. Consequently, for any interval $[a,b]$ and any grid on it of spacing h,*

$$\max_{\nu\in[a,b]} a_T(\nu) \;\le\; \Big(\max_{grid} g + L\,h/2\Big)^{1/2}.$$

The same argument applied to $h(\nu) = \boldsymbol{\phi}_\nu^*\boldsymbol{P}\boldsymbol{\phi}_\nu$ ($\boldsymbol{P} = \boldsymbol{\Phi}(\boldsymbol{\Phi}^*\boldsymbol{\Phi})^{-1}\boldsymbol{\Phi}^*$) gives $\min_{\nu\in[a,b]} v_T(\nu) \ge (1 - (\max_{grid} h + L_P h/2)/N)^{1/2}$.

Proof. $\boldsymbol{\phi}_\nu[j] = e^{i2\pi\nu t_j}$, so $\boldsymbol{\phi}_\nu^* \boldsymbol{M} \boldsymbol{\phi}_\nu = \sum_{j,l} \boldsymbol{M}_{jl} e^{-i2\pi\nu t_j} e^{i2\pi\nu t_l}$, real because $\boldsymbol{M}$ is Hermitian. Its derivative is $g'(\nu) = \sum_{j,l} \boldsymbol{M}_{jl}\, i2\pi(t_l - t_j) e^{i2\pi\nu(t_l - t_j)}$, so $|g'| \le 2\pi \sum_{j,l} |\boldsymbol{M}_{jl}||t_l - t_j| = L$ (a Bernstein-type bound; each exponential is unit-modulus). On a grid of spacing h, any point is within $h/2$ of a grid node, so $g(\nu) \le \max_{\text{grid}} g + L(h/2)$; take the sup and the square root (monotone). The visibility bound is identical with $\boldsymbol{P}$ and $v_T(\nu)^2 = 1 - h(\nu)/N$. □

Everything in $\boldsymbol{M}, \boldsymbol{P}, L$ depends only on the sample times, so the certificate is sample-only; it is validated in `tests/test_design.py::test_certificate_is_sound` (certified bound $\ge$ true value on a fine reference grid, and tight: the Lipschitz slack is negligible at the grid resolutions used). This is exactly the guarantee the finite-candidate bound of Theorem 3(b) cannot give, and for which random designs have no analogue.

7 Additional experimental results

These support the compressed statements in Sec. 6 of the paper. All figures are regenerated from the released `results/*.json` (seeds and environment stamped therein).

Detection (T4). Fig. 1. With calibration/test separation, five sample-only detectors (extended-dictionary ring test, in-sample residual, Lomb–Scargle band test, held-out prediction, cross-fit) track the exact noncentral-χ^2 power curve of T4(b); at the calibrated threshold the test-set false positive rate holds at its nominal 5%. The indistinguishability demonstration instantiates T4(a)'s two-point pair exactly—H1 places a tone at the grid-coherent ν=111, H0 the *same-amplitude* tone at its in-band twin $\nu - Q = -17$, so the grid sample laws are identical—and all five detectors sit at chance (best-of-five AUC 0.51, every bootstrap CI covering 0.5; best TPR at 5% FPR 0.06), at any amplitude. A no-tone null lets energy-sensitive statistics flag that *something* changed, but they cannot attribute it to out-of-band content; that attribution is what is impossible.

Trained-network extension (empirical only). Fig. 2. For Fourier-feature MLPs, SIRENs, and a band-limited head, tone attribution (an ablation: identical initialization trained with and without the tone) is compared to each network's own NTK-linearization prediction—no arbitrary reference band. Over $\ge$ 20 sampling seeds with label-permutation, amplitude, and weight-seed controls, the linear picture extrapolates to *lazy-regime* FF-MLPs (exact fold match 77%, top-3 98%, pattern correlation median 0.96, n=60) and the band-limited head (median 0.98), but *fails for SIRENs* (exact match 20%, correlation median 0.49): trained SIRENs leave their initialization NTK, so the theory makes no claim there. All seed-level failures are released.

Two dimensions. Fig. 3. The linear model reproduces the predicted 2-D frequency-vector fold exactly (planted $(27, 3)$ on a rate-32 grid subset folds to $(-5, 3)$, residual 10^{-15}); trained 2-D networks show the predicted DFT replicas under lattice (coherent) masks (replica-energy ratio 40±32 at the predicted shifts) but not under random masks of equal density (0.99±0.09), and ridge and early stopping do not remove them—structured aliasing is a sampling-design property, not an overfitting artifact.

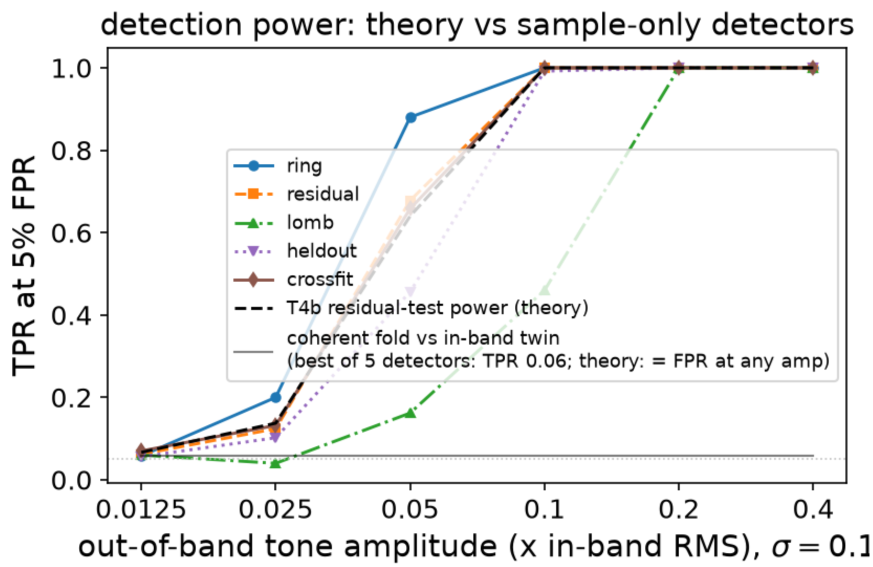

Figure 1: Detection power vs. tone amplitude; the coherent fold sits at chance.

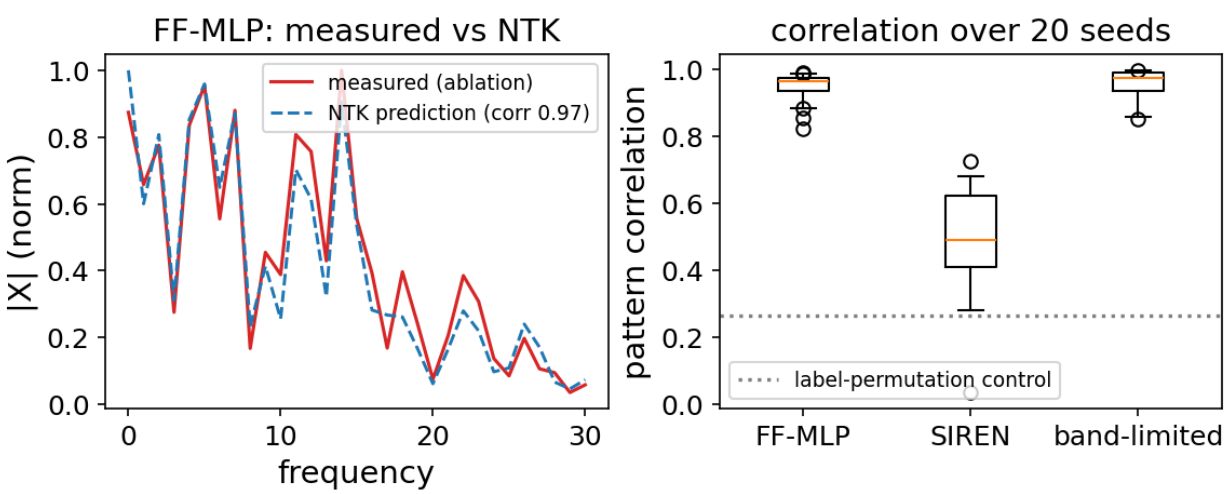

Figure 2: Trained-network tone attribution vs. NTK prediction; SIREN fails.

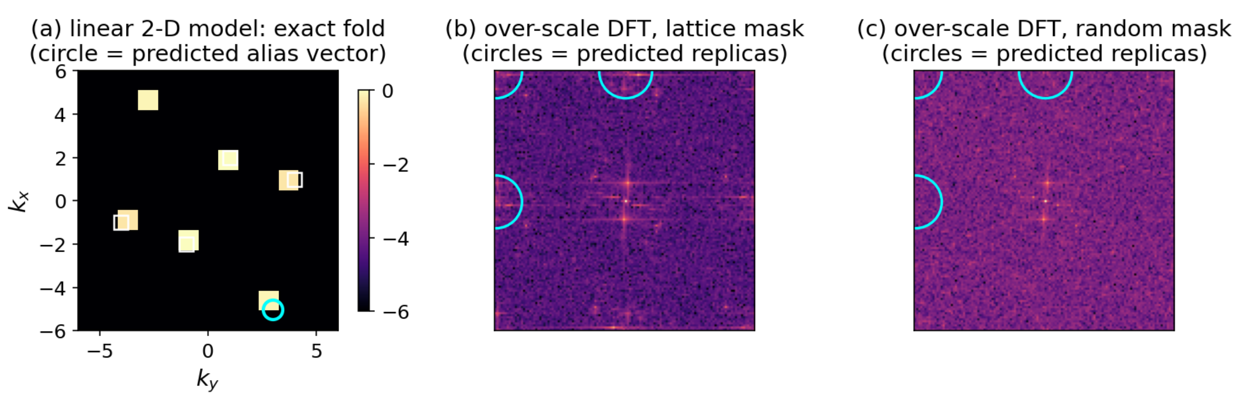

Figure 3: 2-D exact fold (linear) and lattice-vs-random replicas (trained).